

Lin Ding Shan

AgriTech Computer Vision | Edge AI Deployment

Kuala Lumpur, Malaysia · Open to relocation to Australia and New Zealand
l1055505011@gmail.com | linkedin.com/in/lin-ding-shan | github.com/xiaolin200206

PROFESSIONAL SUMMARY

Computer-vision engineer (BSc Data Science, graduating May 2027) who builds vision systems for agriculture and runs them on edge hardware. Built an offline decision-support device for durian disease and pest scouting on Raspberry Pi 5 end to end: field data from 11 orchard sites, detector training, ONNX deployment, quantisation, and latency, power and thermal profiling.

TECHNICAL SKILLS

Languages: Python, SQL

Computer vision: PyTorch, Ultralytics YOLO (v8, 11), RT-DETR, Faster R-CNN, MobileNet, EfficientNet, OpenCV

Edge deployment: Raspberry Pi 5, Linux, ONNX Runtime, LiteRT / TFLite, INT8 and dynamic quantisation, Docker, libcamera

Profiling: latency, power and thermal telemetry, Arm PMU counters, duty-cycle and multi-model scheduling

Data: field data collection, lesion-level annotation, validation on unseen orchards, multi-seed runs

Tools: Git/GitHub, Flask, AWS, Matplotlib, Telegram Bot API

EXPERIENCE

Founder & Lead Engineer | Durian edge-AI scouting device (pre-incorporation agri-tech venture), Malaysia | 2025 – Present

- Built a battery-powered Raspberry Pi 5 decision-support device that helps growers assess 12 durian disease and pest categories from a photo, on demand and fully offline: a field-tested prototype covering image capture, ONNX inference and on-device decoding.
- Built a ~1,300-image field dataset across 11 orchard sites in Peninsular Malaysia and Sabah, setting the lesion-level annotation standard with a crop-protection practitioner; released datasets and code on Zenodo and GitHub.
- Consolidated two specialist detectors into one model with no measurable accuracy loss, cutting operator wait from 718 ms to 351 ms and halving cache-miss traffic.
- Caught INT8 quantisation silently dropping the two smallest pest categories while aggregate accuracy still looked acceptable; switched to weight-only quantisation, keeping FP32-level accuracy at 24–40% lower energy per image.
- Profiled thermal behaviour on-device: traced field interruptions to the 82 °C thermal cut-off and removed them with a 30 s duty-cycle sleep; separately cut peak temperature by 12.7 °C and pest-model latency by 33% by serialising two co-located detectors, with no hardware change.

AI Development Intern | Urban Farm Tech, Malaysia | Jan 2026 – May 2026

- Built and deployed an offline basil disease-monitoring node in a commercial greenhouse: ~1,700 on-site images, MobileNetV2 exported to ONNX on Raspberry Pi 5, Telegram alerts confirmed over 3-of-5 frames, under 2.5 s end to end.
- Profiled native vs Docker execution over four interleaved 3-hour runs with power telemetry: showed the container's apparent power saving vanished per inference, traced its 46% latency gap to an ONNX Runtime version mismatch, and wrote an environment-manifest tool to keep such comparisons attributable.

RESEARCH

Sole-author paper accepted at Smart Agricultural Technology (Elsevier) on thermal cut-off and duty-cycle scheduling for edge inference; six further manuscripts under review at IEEE Transactions and Elsevier journals.

EDUCATION

Bachelor of Computer Science in Data Science with Honours

UCSI University, Kuala Lumpur, Malaysia | Sep 2024 – May 2027 (expected)

Languages: English (fluent), Chinese (native), Malay (beginner)